Zag Farkle - Homework #3 Summary

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Project Goal: Extent the object-oriented Farkle game from Homework #1 and 2 by introducing full house hands, play to specified point amount, multiplayer functionality, score outputs, and an extended unit testing file.

Design Overview

This application stores components of the game Farkle across multiple Java classes. Farkle.java serves as the entry point, while FarkleGame.java manages the interactive state, user prompting, and the turn loop. Hand.java holds multiple Die.java objects, and FarkleScorer.java consists of the scoring rules. FarkleTester.java tests valid hands.

Starting Screen

```
************************************************************************
*                   Zag Farkle by Darrell Cenido!                      *
*                         Copyright: 2026                              *
************************************************************************

How many points would you like to play in this game? [10000]
How many players are going to be playing? [1]
What is the player #1's name? [Unknown player #1]
--------------------------------

--- Round 1: Unknown player #1's Turn ---
Unknown player #1, it is your turn! Rolling dice...
```

Full House Combo

```
****************** Current hand and meld ******************

 Die   Hand |   Meld
------------+---------------
 (A)    1   |
 (B)    3   |
 (C)    3   |
 (D)    3   |
 (E)    5   |
 (F)    5   |
------------+---------------
               Meld Score: 0
               Turn Points: 0

 (K) Bank Meld & End Round
 (R) Reroll unmelded dice
 (Q) Quit game

Enter letters for your choice(s): bcdef

****************** Current hand and meld ******************

 Die   Hand |   Meld
------------+---------------
 (A)    1   |
 (B)        |     3
 (C)        |     3
 (D)        |     3
 (E)        |     5
 (F)        |     5
------------+---------------
               Meld Score: 1500
               Turn Points: 1500
```

Winning Single Player

```
Current total points:
Darrell 1800

********************************************************************************
 CONGRATULATIONS DARRELL! YOU WON!
 Final Scores:
 Darrell        1800
********************************************************************************
```

Entering Multiple Players

```
********************************************************************************
*                    Zag Farkle by Darrell Cenido!                             *
*                           Copyright: 2026                                    *
********************************************************************************

How many points would you like to play in this game? [10000] 1000
How many players are going to be playing? [1] 2
What is the player #1's name? [Unknown player #1] Darrell
What is the player #2's name? [Unknown player #2] Danielle
```

Total Scores

```
Current total points:
Darrell 100
Danielle        1250
```

Winner's Screen

```
********************************************************************************
 CONGRATULATIONS DANIELLE! YOU WON!
 Final Scores:
 Darrell        100
 Danielle       1250
********************************************************************************
```

Unit Tests

In FarkleTester.java, I tested if the full house is correctly scored by feeding the frequency array [0, 0, 2, 0, 3, 0, 0] and [0, 3, 2, 0, 0, 0, 0] into FarkleScorer.calculateMeldScore() and checking that the returned value is 1500 points.

Issues

The part that was the most difficult was making sure that the default names worked correctly. It wasn't necessarily difficult to achieve it was just the most time consuming portion of the assignment.

Retrospective

For this assignment I wish I would have started it earlier it just simply was not possible given the amount of work that I had to complete before this for other classes.